

iSPEC 2026

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IEEE PES KERALA
CHAPTER
4-6 December, 2026



The 7th edition of iSPEC Conference

Trivandrum, Kerala, India | 4-6 December 2026



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Engineering ,Trivandrum.

The IEEE Sustainable Power and Energy Conference (iSPEC) 2026 is a premier international forum that brings together researchers, engineers, policymakers, and industry leaders from across the globe to present and discuss the latest advances in power systems and energy technologies.

Paper Submission

- Full Paper Deadline: 30 June 2026
- Acceptance Notification:
30 September 2026
- Final Paper Submission: 15 October 2026

Registration

- Registration Deadline:
1 November 2026

Conference

- Conference Dates:
4 – 6 December 2026

Submission : <https://ispec2026.org/submission>

Registration : <https://ispec2026.org>

contact : contact@ispec2026.org

"Integrated Pathways in Sustainable Power and Energy for Carbon Neutrality"

The IEEE Sustainable Power and Energy Conference (iSPEC) 2026 will be held at MBCET Trivandrum, Kerala under the theme "Integrated Pathways in Sustainable Power and Energy for Carbon Neutrality"

The conference brings together researchers, industry professionals, utilities, policymakers, and educators to exchange ideas and showcase advances in power and energy systems, with a focus on sustainability, digitalization, resilience, and system-level innovation.

The technical program will include keynote speeches, panel sessions, tutorials, and peer-reviewed paper and poster presentations, highlighting both research breakthroughs and practical applications. With eight technical tracks delivered over three days, iSPEC 2026 provides a dynamic platform for knowledge sharing, collaboration, and professional development.

Conference Topics

Track 1: Renewable Energy & Sustainable Technologies

- Advanced solar PV modeling, forecasting, and performance optimization
- Wind energy conversion systems
- hybrid energy systems
- renewable integration
- Green hydrogen and fuel cells
- Biomass, waste-to-energy, and circular energy systems
- Sustainability assessment, carbon neutrality, and life cycle analysis

Track 2: Grid Infrastructure & Resilience

- Modern transmission and distribution system planning
- Microgrids and resilient distributed energy architectures
- Grid hardening strategies against extreme weather events
- Black start capability and system restoration techniques
- Wide-area monitoring systems (WAMS) and synchrophasor applications
- Power quality issues and solutions

Track 3: Protection, Control & Automation for Power Grid Operation

- Adaptive and AI-based protection schemes
- Wide-area protection and coordinated control
- Substation automation and IEC 61850 applications
- Fault detection, isolation, and service restoration (FDIR)
- HVDC and FACTS-based control strategies



Track 4: Smart Grids, AI & Digitalization

- AI/ML applications in load forecasting and stability assessment
- Digital twins for power systems
- DER Integration and IoT-enabled smart energy management systems
- Cybersecurity in smart grid infrastructure
- Big data analytics and cloud computing in utilities
- HIL /SIL based simulation systems



Track 5: Transportation Electrification

- Electric vehicle (EV) charging infrastructure and standards
- Vehicle-to-Grid (V2G) and bidirectional charging technologies
- Battery management systems (BMS) and state estimation techniques
- Electrification of public and freight transportation
- Power electronics for EV propulsion systems

Track 6: Energy Storage Systems

- Advanced battery technologies and solid-state batteries
- Supercapacitors and hybrid storage systems
- Grid-scale storage integration and control
- Thermal energy storage systems
- Degradation modeling and lifetime prediction of storage systems
- Design and siting of energy storage systems

Conference Topics

Track 7: Energy Markets, Policy & Governance

- Peer-to-Peer (P2P) Energy Trading and Decentralized Energy Markets
- Transactive Energy Systems and Blockchain Applications
- Regulatory and Policy Frameworks for Distributed Energy Resources and Energy Trading
- Market Design, Demand Response Programs, and Governance
- Models for Future Energy Systems



Track 8: Advanced Power Converters and Electric Drives for Sustainable Energy Systems

- High-Efficient Power Converters for Renewable Integration
- Electric Drives for Sustainable Transportation and Industrial Electrification
- Grid-Interactive Power Electronics and Energy Storage Systems
- Advanced Modulation, Control, and AI-Based Optimization in Power Electronics

